01 PAGE / 08 — STACKCALC SC-32 SCIENTIFIC RPN CALCULATOR

4-Register Live Stack Display, Tactile Diamond Keys & Origami Stand

01 02 03 04

SKU: SC-32-HW • KEEP IN LID

SIDE A • COVER (TOP-LEFT)

WHAT IS THIS BOX FOR? THERE IS NO EQUALS BUTTON (ON PURPOSE).

Don't panic! Once you try Reverse Polish Notation (RPN) on a live 4-level stack, ordinary calculators feel like typing with oven mitts.

WELCOME TO THE 4-REGISTER RPN STACK

How do you calculate 3 + 4 without an equals button? Just stack them:

- **X-Register (Bottom Line): Where you type new numbers and see results**
- **Y-Register (Line 3): Press [ENTER] to push your first number up here**
- **Z & T Registers (Top Lines): Hold earlier steps while you work**

Try this right now: Press [3] [ENTER] [4] [+]. Boom—7 appears instantly, with zero parentheses and zero order-of-operations drama!

FOLDING ORIGAMI DESK STAND (25° & 65°)

Four interlocking puzzle trusses slide together into a sturdy desk cradle (choose 25° typing tilt or 65° display mode), then pack flat for travel.

WHAT'S IN YOUR BOX (11 PIECES)

- 1x SC-32 Handheld RPN Calculator (with 35 Tactile Diamond Keys)
- 1x 4-Piece Interlocking Desk Stand + 1x Protective Travel Sleeve
- 1x Stainless Hex Screw Kit + 1.5mm L-Hex Key (uses 1x CR2032 coin cell)

★ YOUR FIRST 60 SECONDS WITH THIS BOX — START HERE!

1. **COMPUTE 3 + 4 IN RPN:**
   Press 3 [ENTER] 4 [+]. With no '=' key, pressing [+] combines X and Y immediately!
2. **SNAP TOGETHER THE ORIGAMI STAND:**
   Slot the 2 crossbars into the 2 side trusses for a 25° typing tilt (or flip for 65° display mode).
3. **EXPLORE [↱][POLY], [↱][SOLVE] & [BASE]:**
   Press [↱][POLY] to solve quadratics/cubics, [↱][SOLVE] for numerical roots, or [BASE] for HEX/BIN!

EQN **EVERY MANIPULATIVE BOX ICON MAPS DIRECTLY TO YOUR SC-32 KEYPAD!**

The button icon embossed on your box lid (left) is the exact calculator key ([ENTER], [↰][x²], [1/x] & [↱][POLY])! Build every lesson with your blocks, then press the key.

4-LEVEL RPN STACK ENGINE ↔ (4 + 5) × (6 + 7) = 117

STEP A: HANDS-ON WITH SC-32 KEYPAD — 1:1 CAD SCENE



25° Bench Tilt — Ergonomic Key Entry

65° Upright Mode — Desk Reference Display

9-Tap RPN Tree — Zero Parentheses

▲ Every manipulative box button icon in the StackCalc lab corresponds directly to an operation on the SC-32!

EQN STEP B: TRY IT ON YOUR CALCULATOR — BOX ICON: [ENTER], [↰][x²],

RPN EXPRESSION TREE: (4 + 5) × (6 + 7) = 117

9 × 13 = 117

Y: 4 + 5 = 9 | X: 6 + 7 = 13

[ENTER] | [+] → 9 | [ENTER] | [+] → 13

4 [ENTER] 5 [+] 6 [ENTER] 7 [+] [×] → 117

PRESS BOX ICON & RPN KEYS (SC32):

1. FIRST SUM: 4 ENTER 5 + = 9
2. 2ND & ×: 6 ENTER 7 + × = 117

✓ DISPLAY (X) = 117

KEYSTROKES: 4 [ENTER] 5 [+] 6 [ENTER] 7 [+] [×]

Evaluate nested expressions from the inside out: both sub-results sit visibly in Y (9) and X (13) before [×]!

WELCOME TO HANDS-ON MATH • KEEP THIS GUIDE IN YOUR BOX LID

STEP 1: UNFOLD RIGHT → PAGE 02

▲ FOLD 1: FOLD BOTTOM HALF BEHIND TOP ▲

▼ UNFOLD STEP 2: OPEN DOWN FOR PAGES 03 & 04 ▼

02 PAGE / 08 — HOW IT WORKS: THE CORE IDEA

4-Level RPN Stack (X, Y, Z, T): Compute Without Parentheses

01 02 03 04

SKU: SC-32-HW • KEEP IN LID

SIDE A • 1ST UNFOLD (TOP-RIGHT)

POSTFIX STACK MACHINE & ALGEBRAIC PURITY

In infix notation, expressions like (A + B) × (C + D) require parentheses because operator precedence must be overridden.

In postfix Reverse Polish Notation (RPN), operators act immediately:

A ENTER B + C ENTER D + ×

The automatic 4-level stack maintains operands in physical order:

- **Stack Lift: Entering a number pushes X → Y → Z → T**
- **Binary Operator: Pops X and Y, executes f(Y, X), drops Z → Y, T → Z**

Zero keystrokes are wasted on parentheses. Intermediate results remain accessible for inspection without retyping or losing context!

MONOLITHIC RPNCore ENGINE FIRMWARE

The calculator is powered by StackCalc's open-source RPNCore engine, written in pure embedded Swift and C. Identical engine code runs across the physical hardware, Apple Watch target, iOS application, and cross-surface fuzzing test harnesses.

CR2032 COIN-CELL BATTERY PROTOCOL

Due to international air freight and postal restrictions on lithium cells, the CR2032 coin battery is excluded from mailings. Insert one standard 3V CR2032 battery into the gold chassis terminal with '+' positive side UP before powering on.

3 CLASSIC MATH TRAPS THIS KIT MAKES IMPOSSIBLE — VISUAL PROOF

✗ **TRAP 1 — Pressing [ENTER] after the 2nd number:**
✓ **FIX:** Only press [ENTER] to separate TWO numbers typed in a row: 3 [ENTER] 4 [+] (not 4 [ENTER] [+])!!

✗ **TRAP 2 — Pressing [ENTER] before [↰][x²] or [√x]:**
✓ **FIX:** Single-number keys ([↰][x²], [√x], [1/x], [SIN]) act immediately on X without needing [ENTER]!

✗ **TRAP 3 — Clearing the whole stack on a typo:**
✓ **FIX:** Use [←] while typing, or [↰][CLx] to clear only Register X while keeping Y, Z, and T intact!

CALCULATOR CHECK: 4 [ENTER] 5 [+] 6 [ENTER] 7 [+] [×] → X = 117

4-LEVEL STACK REGISTERS

REGISTER	ROLE	STACK BEHAVIOR	DISPLAY
T (Top)	Constant Hold	Replicates on drop	Top LCD Line
Z (Third)	Chained Storage	Drops to Y	Line 2
Y (Second)	Operand 1	Pushed by [ENTER]	Line 3
X (Bottom)	Active / Result	Receives keyboard	Main Line

HARDWARE ASSEMBLY & TORQUE SPECIFICATION

- Fasteners: 4x M2 × 8.0 mm 304 Stainless Steel socket head cap screws.
- Threaded Inserts: M2 brass heat-set inserts pass 1.2 N·m torque retention check.
- Hex Tool: 1.5mm L-shape hex key included for easy battery hatch access.
- Tactile Membrane: Semi-clear flexible silicone with 1.8mm travel stroke.
- LCD Display: High-contrast transflective screen readable in direct sunlight.
- Drop Protection: Reinforced corners tested against 1.5-meter concrete drops.

1:1 CAD REFERENCE VIEW — BOX ICON & FORMULA QUICK-REFERENCE

1:1 BLOCK CAD ANATOMY

SOCKET (1.85mm) ▲ 1.85m — 1U = 7.00 mm VISIBLE DATUM — 7mm — BASE PLUG (1.85mm) ▼ 1.85m

✓ FLUSH PLUG-AND-SOCKET JOINT
Socket swallows plug completely:
Only 7.00mm true heights add!

Push & Separate:
A [ENTER] B (A in Y, B in X)

Square Any Number:
N [↰] [x²] or N [ENTER] [×]

Swap or Roll Stack:
[xey] swaps X & Y | [R↓] rolls stack

Undo / Recover Last X:
[↱] [LASTx] recalls previous X before op

Built-In Solvers:
[↱][POLY] | [↱][SOLVE] | [↱][∫] | [TVM]

REVERSE POLISH NOTATION • SEE EVERY INTERMEDIATE STEP LIVE

STEP 2: UNFOLD DOWN ▼ PAGES 03–04

03 PAGE / 08 — EXAMPLE 1: HANDS-ON WALKTHROUGH

Pythagorean Hypotenuse on the Stack: √(3² + 4²) = 5

01 02 03 04

SKU: SC-32-HW • KEEP IN LID

SIDE A • 2ND UNFOLD (BOT-LEFT)

1 PYTHAGOREAN HYPOTENUSE & 4-REGISTER STACK

THE PROBLEM: Compute √(3² + 4²) = √(9 + 16) = √25 = 5.0000.

1. SQUARE THE FIRST LEG (3²):
Press [3] [x²] → calculates 3² = 9 into X-register.

2. SQUARE THE SECOND LEG (4²):
Press [4] [x²] → 9 automatically rises to Register Y!
Register X now displays 4² = 16.0000.

3. SUM LEGS & TAKE SQUARE ROOT:
Press [+] → adds Y (9) and X (16); displays 25 in X.
Press [√x] → takes square root of 25.
→ Display shows 5.0000 in Register X!

ZERO PARENTHESES • ZERO MEMORY STORES

Notice that you never had to press "(" or ")", and you never had to write down or store intermediate results. Register Y automatically preserved 9 while you computed 4², and [+] collapsed both registers instantaneously!

EQN STEP B: TRY EXAMPLE #1 ON YOUR CALCULATOR — [ENTER], [↰][x²],

STEP 1 (SQUARE 3):
Press 3 [↰] [x²] → Register X immediately becomes 9.

STEP 2 (SQUARE 4 & ADD):
Type 4 (auto-lifts 9 into Y!), press [↰] [x²] (= 16 in X), then [+] → X = 25.

STEP 3 (SQUARE ROOT):
Press [√x] → X = 5 in 6 clean keystrokes with zero parentheses!

LIVE 4-REGISTER RPN STACK READOUT (AFTER KEYSTROKES):
T: 0 | Z: 0 | Y: 0 | X: 5 → DISPLAY (X) = 5

LESSON 1: 4-REGISTER STACK CHASSIS ↔ HYPOTENUSE KEYSTROK

STEP A: HANDS-ON WITH SC-32 HARDWARE & STAND — 1:1 CAD SCENE



4 Live Registers — T, Z, Y, X Simultaneous

43 Tactile Keys — Primary, [↰] Gold, [↱]

RPN Architecture — Zero Parentheses Needed

▲ Unlike 1-line algebraic calculators that hide intermediate numbers, the SC-32 shows all 4 stack levels (T, Z, Y, X) live!

EQN STEP B: TRY IT ON YOUR CALCULATOR — BOX ICON: [ENTER], [↰][x²],

HYPOTENUSE TREE: √(3² + 4²) = 5

√25 = 5

3 [↰][x²] = 9 | 4 [↰][x²] = 16

Y = 9 | X = 16 | [+][√x] = 5

3 [↰][x²] 4 [↰][x²] [+] [√x] → 5

PRESS BOX ICON & RPN KEYS (SC32):

1. SQUARES: 3 ↰ x² 4 ↰ x² +
2. ROOT: √x = 5

✓ DISPLAY (X) = 5

KEYSTROKES: 3 [↰] [x²] 4 [↰] [x²] [+] [√x]

Because [↰][x²] acts immediately on X, typing 4 automatically lifts 9 into Register Y—zero parentheses!

✓ 4-REGISTER STACK: T: 0 | Z: 0 | Y: 0 | X: 5 → X = 5

NO EQUALS BUTTON NEEDED • OPERATORS ACT INSTANTLY ON X AND Y

CONTINUE RIGHT → PAGE 04

04 PAGE / 08 — EXAMPLE 2: CORE MATHEMATICAL PRINCIPLE

Why Physical Block Towers and the RPN Stack Are Identical

01 02 03 04

SKU: SC-32-HW • KEEP IN LID

SIDE A • 2ND UNFOLD (BOT-RIGHT)

2 THE PHYSICAL-TO-DIGITAL DUALITY

THE CORE PRINCIPLE: Why is the calculator called "StackCalc"?

1. LOOK AT YOUR KINESTHETIC TOWERS:
In the Prime Factor Studio or Fraction Studio, you physically stack blocks on top of one another to build composite numbers.

2. THE PHYSICAL-DIGITAL 1:1 MAPPING:
- Placing a block onto the tower = Pressing [ENTER] (Stack Lift)
- Snapping two towers together = Pressing [+] or [×] (Operator evaluation)
- Reading the tower summit = Looking at the X-register display

3. COMPLETE CONTINUUM:
The physical manipulative gives the spatial intuition.
The SC-32 calculator gives high-speed scientific execution.

✓ KINESTHETIC ROOTS OF COMPUTATION

RPN is not just a keystroke shortcut.

It is the natural physical grammar of how humans build with blocks!

By using the SC-32 alongside StackCalc physical manipulatives, students bridge hands-on tactile play into professional engineering **fluency without abstract gaps.**

★ 2-PLAYER GAME: 'NO-PARENTHESIS SPEED DUEL' (2 MINS)

Write a nested fraction like ((3 + 7) × (9 − 2)) ÷ (4 + 1) = 14 on a sticky note. Player A uses a phone's standard algebraic calculator; Player B uses the SC-32 in RPN (3 [ENTER] 7 [+] 9 [ENTER] 2 [−] [×] 4 [ENTER] 1 [+] [÷]). **Watch RPN win with fewer keystrokes!**

⚙ WHERE THIS SHOWS UP IN COMPILERS, CPU STACKS & POSTSCRIPT

Every modern CPU, WebAssembly virtual machine, and Python bytecode interpreter evaluates math using a stack in Reverse Polish Notation! When you master the SC-32, you are literally thinking **the way computer compilers and Hewlett-Packard engineering instruments think.**

LESSON 2: MODULAR ARCHITECTURE ↔ [1/x] HARMONIC RECIPROC

STEP A: HANDS-ON WITH MODULAR ARCHITECTURE — 1:1 CAD SCENE



Origami Stand — 25° Typing / 65° Displa

Dovetail Battery Bay — Tool-Free Slide Release

6-Tap Reciprocal — 3 [1/x] 4 [1/x] [+] [1/

▲ Engineered for repairability and daily desk use: evaluate parallel reciprocals in 6 taps with [1/x]!

EQN STEP B: TRY IT ON YOUR CALCULATOR — BOX ICON: [ENTER], [↰][x²],

HARMONIC RECIPROCAL TREE: 1/(1/3 + 1/4)

12/7 = 1.7143

1/3 + 1/4 = 7/12 | Invert [1/x]

3 [1/x] | 4 [1/x] | [+] [1/x]

3 [1/x] 4 [1/x] [+] [1/x] → 1.7142857143

PRESS BOX ICON & RPN KEYS (SC32):

1. INVERT: 3 1/x 4 1/x +
2. TOTAL R: 1/x = 1.7143

✓ DISPLAY (X) = 1.7142857143

KEYSTROKES: 3 [1/x] 4 [1/x] [+] [1/x]

Matches your FRAC-01 blocks (1/3 + 1/4 = 7/12) and inverts into 12/7 in 6 clean keystrokes!

✓ 4-REGISTER STACK: T: 0 | Z: 0 | Y: 0 | X: 1.7142857143 → X = 1.7142857143

THE HEART OF STACKCALC • CONNECTS EVERY KIT IN THE LAB

STEP 3: FLIP SHEET OVER ↻ PAGE 05

05 — EXAMPLE 3 & DESK CHALLENGES

PAGE / 08

RPN Expression Tree: (4 + 5) × (6 + 7) = 117 + 3 Challenges

3 RPN EXPRESSION TREE: (4 + 5) × (6 + 7) = 117

THE GOAL: Compute (4 + 5) × (6 + 7) in RPN with zero parenthesis keys!

1. COMPUTE FIRST TERM (4 + 5):
Press [4] [ENTER] [5] [+]. Register X now displays 9.0000.

2. COMPUTE SECOND TERM (6 + 7):
Press [6] [ENTER] [7] [+]. 9 rises to Y; Register X displays 13.0000.

3. MULTIPLY THE SUB-TREES:
Press [×]. Registers collapse: 9 × 13 = 117.0000 displays in X!

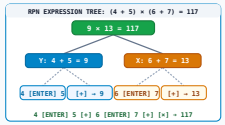
✓ ZERO PARENTHESES NEEDED: THE RPN STACK SECRET

In standard calculators, nested parentheses (4+5)×(6+7) cause syntax errors.
In RPN, intermediate result 9 waits safely in Register Y while you compute 6 + 7.
Pressing [×] multiplies Y and X directly—matching an evaluation tree!
Verify on calculator: 4 [ENTER] 5 [+] 6 [ENTER] 7 [+] [×] delivers 117.0000!

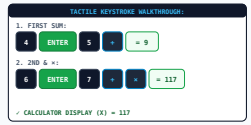
PRO TIP: CLOSE YOUR EYES & FEEL THE ANSWER

Once you build an example, run your fingers across the pieces. Your hands remember physical shapes and heights way longer than memorized rules!

EXAMPLE 3: VISUAL DECOMPOSITION TREE



STEP B: PRESS BOX ICON & RPN KEYS (SC32)



EQN

FULL KEYSTROKES: 4 [ENTER] 5 [+] 6 [ENTER] 7 [+] [×]
✓ 4-REGISTER STACK: T: 0 | Z: 0 | Y: 0 | X: 117 → X = 117

3 DESK CHALLENGES — TRY THESE NOW!

WARM-UP #1 — **Compute 5 × (3 + 8) in RPN!**
Step A: Build with blocks on your Lid Tray • Step B: Verify on SC-32 Calculator
✓ Build first, then flip booklet upside-down to check Answer #1 below!

LEVEL-UP #2 — **Compute (12 × 18) ÷ 6 in one smooth chain!**
Step A: Build with blocks on your Lid Tray • Step B: Verify on SC-32 Calculator
✓ Build first, then flip booklet upside-down to check Answer #2 below!

BRAIN-MELTER #3 — **Find the harmonic sum 1/3 + 1/4 using [1/x]!**
Step A: Build with blocks on your Lid Tray • Step B: Verify on SC-32 Calculator
✓ Build first, then flip booklet upside-down to check Answer #3 below!

SELF-CHECK ANSWER KEY (ROTATE BOOKLET 180° TO READ)

(1/Z1/) 0.5833 = [+] [X/Y] [÷] [X/Y] [C] :#E

#1: [ENTER] [8] [ENTER] [12] [+] [5] = [÷] [6] = 5.0000 | #2: [12] [ENTER] [18] [×] = 36.0000

FOR PARENTS & TEACHERS (Middle School, High School, College & Engineering)

- 4-Register Live RPN Stack (T, Z, Y, X visible simultaneously on screen).
- Full Scientific Functions: sin, cos, tan, asin, acos, atan, ln, log, e^x, y^x, √x, 1/x.
- Powered by open-source RPNCore engine (identical across hardware, iOS & watchOS).
- **CCSS.MATH.PRACTICE.MP2, MP4, MP5, MP7: Reason, Model, Use Tools & Structure.**

Next up on Page 06 → How to stack everything back in the box in 10 seconds!

BOSS CHALLENGE #4 — **BLOCKS → BOX ICON KEY SPEED-RUN!**

1. Build Examples #1, #2, and #3 hands-on with your physical pieces.
2. Press the Box Button Icon ([ENTER], [↑][x²], [1/x] & [r][POLY]) and keystrokes on your calculator:
   • 4 [ENTER] 5 [+] 6 [ENTER] 7 [+] [×] → X = 117

CHECK OFF YOUR MASTERY MILESTONES:

- □ 1. Demonstrated Lessons #1, #2 & #3 hands-on with the physical blocks
- □ 2. Solved all 3 Desk Challenges above (check upside-down key if stuck!)
- □ 3. Demonstrated each lesson on the calculator using the Box Button Icon
- □ 4. Packed all pieces back into the box in under 10 seconds (see Page 06 →)

TEST YOURSELF • ANSWERS INCLUDED SO YOU CAN CHECK RIGHT AWAY

NEXT: HOW TO PACK UP → PAGE 06

06 — HOW TO STACK IT ALL BACK IN

PAGE / 08

The 10-Second Self-Checking Pack-Up Puzzle

TOP-DOWN BOX MAP: WHERE EVERY PIECE LIVES

Putting the blocks away is a 10-second puzzle. Match each piece to its labeled home bay below so everything nests flush and the lid snaps shut:



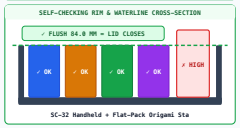
✓ PACKING THE CALCULATOR & FOLD-FLAT ORIGAMI STAND

Slide the SC-32 calculator face-up into its protective travel sleeve.
To pack the 4-piece Origami Desk Stand, unclip the 2 crossbars from the 2 side trusses—all 4 flat pieces tuck into the rear pocket of the sleeve.
Keep the 1.5mm L-Hex key in its slot for quick CR2032 coin-cell battery changes!

1. STORAGE CASE & COMPARTMENTS



2. WHY THE LID SELF-CHECKS EVERY PIECE



▲ SELF-INDEXING BAYS & EMBOSSED BOX BUTTON ICON:
Calibrated bays guide every piece home, and the lid icon links directly to your calculator.

THE 10-SECOND PACK-UP ROUTINE

HOW TO STACK EVERYTHING BACK IN:

1. SORT BY COLOR & NUMBER:
Group matching blocks or tiles together on your desk first.

2. STACK & DROP INTO LABELED BAYS:
Follow the Top-Down Box Map on the left. If a tower looks too tall or too short, a block is in the wrong bay—swap it until everything levels!

3. TUCK THIS CARD IN THE LID & SNAP SHUT:
Fold this guide back to [01] and rest it right on top before closing.

DUAL-ANGLE DESK STAND (25° & 65°) & BATTERY TIP

- Slot the 4 stand pieces together one way for a 25° typing tilt, or flip for a 65° display cradle.
- Uses 1x standard CR2032 3V coin cell (insert with '+' positive side facing UP).
- Keep this folded guide in your box or sleeve pocket for RPN shortcuts & Part IDs!

SHORT A PIECE AT CLEAN-UP TIME?

First check under notebooks and chair legs! If a piece truly vanished, **see PAGE 08 (bottom-right) for your complete Part ID Reference Table.**

✓ SC-32-HW 4-POINT CLEAN-UP CHECKLIST

1. DISASSEMBLE STAND:	Unslot the 2 crossbars from the 2 side trusses so all 4 pieces lie flat.
2. SEAT IN CRADLE:	Place the SC-32 calculator and flat stand trusses into their bays.
3. FOLD CARD TO [01]:	Fold this reference booklet to Panel [01] and rest on top.
4. SLIDE INTO POUCH:	Slip the calculator and stand into the TPU protective travel sleeve!

MODULAR CLASSROOM & BOOKSHELF NESTING (42.0 MM PITCH)

Every StackCalc box is sized in exact multiples of 42.0 mm (1×1, 3×1, 3×2). All 9 kit boxes interlock side-by-side in a classroom cart or drawer with zero wasted space, and every embossed box icon matches its calculator operation key!

TRAVEL SLEEVE & FOLD-FLAT STAND • READY FOR DESK OR BACKPACK

NEXT: CONNECTED LAB ▼ PAGE 07

SIDE B (PAGES 05–08) • READ TOP ROW FIRST

THEN READ BOTTOM ROW (PAGE 07 → PAGE 08)

07 — THE STACKCALC CONNECTED LAB

PAGE / 08

How Your Kits & RPN Calculator Stack Work Together

ONE CONNECTED SYSTEM FROM KINDERGARTEN TO CALCULUS

Every StackCalc box uses the same modular grid and unit heights—so pieces from different kits actually fit and work together.

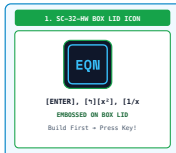
WHERE YOUR BOX FITS IN THE 9-KIT LAB

BASE-10 **Ten-Frames Lab** Early Counting & Regrouping	DOMI-01 **Math Domino Studio** Doubles & 12-Frame Part-Whole	FRAC-01 **Fraction Studio** Proportional Towers & 4-Way Faces
PRIME-01 **Prime Factor Studio** Multiply by Stacking Prime Towers	ALGB-01 **Algebra Studio** Quadratic Area Arrays & Factoring	TIME-01 **Planetary Clock** 12:1 Geared Hands & Time Wedges
PROB-01 **Probability Dice** Fair Polyhedrals & 10-Sector Spinner	VOLV-01 **Unit Circle Volvelle** Spin-to-Read Exact Trig Radicals	SC-32-HW ◂ YOUR BOX **SC-32 RPN Calculator** 4-Register Stack Scientific Engine

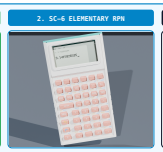
WHY THIS MATTERS: ZERO "THROW-AWAY" MATH

In most classrooms, every grade uses a totally unrelated plastic toy.
In StackCalc, the 7mm unit block you count with in 1st grade is the exact same height as the 1/12th fraction block and the [2] prime block later on!

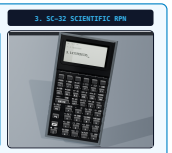
1. SC-32-HW BOX LID ICON



2. SC-6 ELEMENTARY RPN



3. SC-32 SCIENTIFIC RPN



▲ BOX ICON = CALCULATOR KEY CONTINUITY:
Every physical box lid carries the exact icon of the calculator key that computes that lesson!

HOW BLOCKS CONNECT TO THE CALCULATOR

SHARED MODULAR FOOTPRINT & HEIGHTS

- Docks alongside any StackCalc kit box on your desk using the shared 42mm modular footprint.
- Recessed sockets swallow the bottom peg so only true heights add.
- All storage boxes nest neatly side-by-side on any desk or shelf.

WHY WE CALL IT "STACKCALC" (THE RPN SECRET)

- The digital brain of the StackCalc lab: every physical block stack translates 1:1 into RPN keystrokes.
- Placing a block on the table = pressing [ENTER] to push onto the stack.
- Snapping two towers together = pressing [+] or [×] with zero parentheses!
- Mirrors the 4 stack levels on the SC-32 calculator: X, Y, Z, and T.

FREE ACTIVITY DECKS & INTERACTIVE SIMULATORS

Download extra printable puzzle cards, teacher lesson guides, and interactive browser simulations anytime—no login needed.

Explore All 9 Kits & Free Guides at stackcalc.io

LIVE 4-REGISTER RPN STACK

T: 0 (Top Memory)
Z: 0 (3rd Level)
Y: 0 (1st Block)
X: 5 (√(3² + 4²)) (Active Display)

HOW BLOCKS MAP TO STACK KEYS

1. PLACE 1ST PIECE ON STAGE:
Type number + [ENTER] → lifts to Y
2. ADD / STACK 2ND PIECE:
Type 2nd number in X, then [+]/[×]
3. PRESS BOX ICON KEY:
Press [ENTER], [↑][x²], [1/x
✓ Zero Parentheses • Always Visible!

EQN

BOX ICON = [ENTER], [↑][x²], [1/x] & [r][POLY] • VERIFIED LESSON SEQUENCES:
• Lesson 1: 3 [↑] [x²] 4 [↑] [x²] [+] [√x] → X = 5
• Lesson 2: 3 [1/x] 4 [1/x] [+] [1/x] → X = 1.71428571

MODULAR 9-KIT CONTINUUM • FROM EARLY COUNTING TO AP CALCULUS

NEXT: PART ID TABLE → PAGE 08

08 — KEEP THIS CARD! PART IDs & VIDEO

PAGE / 08

Complete 11-Piece Reference & 60-Second 3D Video Guide

PART ID REFERENCE TABLE (11 TOTAL PIECES)

✓ KEEP THIS CARD INSIDE YOUR BOX LID!
Backpacks eat blocks. Dogs chew dice. Vacuum cleaners take no prisoners. Keep this card in your lid so you always have the exact Part IDs below for single-piece refills!

PART NAME	QTY	COLOR / FINISH	PART ID
Tapered Calculator Chassis	1x	Black	SC-CHAS-7f9
Top Bezel Cowl Cap	1x	Black	SC-TOP-1b5
Unified Keycap Faceplate	1x	Gold / Black	SC-UNIF-fa4
Tactile Key Membrane	1x	Clear	SC-MEM-4fc
Origami Stand Trusses	4x	Black	SC-PUZZ-3f1
Protective Travel Sleeve	1x	Gray	ACC-SLV-9a0
M2x8mm Stainless Screws	4x	Silver	SC-FAST-001
1.5mm L-Hex Key & Feet	1x	Silver	SC-TOOL-001

NEED A REPLACEMENT PIECE? NO NEED TO BUY A WHOLE NEW BOX.
Enter any Part ID above at stackcalc.io/parts to order individual replacement pieces.

COLOR & PIECE IDENTIFICATION KEY — MATCH ANY LOOSE PART

Black	Gold / Black	Clear	Silver
Ergonomic Calculator Body	35 Diamond Keycaps	Tactile Key Membrane	M2 Socket Head Screws

UNIVERSAL STACKCALC HEIGHT PITCH (1U = 7.00 MM • 12U = 84.00 MM ONE WHOLE)
Measure any loose piece in 7mm units (U) against this scale bar to identify it:



✓ DISTRICT PO & SAFETY: 10-Pack SKU SC-32-HW-CLASS10 • CPSIA / ASTM F963 Non-Toxic Washable PLA/ABS

WATCH THE 60-SECOND 3D VIDEO GUIDE

POINT CAMERA TO WATCH DEMO
stackcalc.io/products/sc32-calculator/
See animated 3D examples & pack-up walkthrough

QUESTIONS, CLASSROOM SETS & SUPPORT

- Email us anytime: support@stackcalc.io
- School & District POs: school-orders@stackcalc.io

BATTERY NOTE: Uses 1x CR2032 3V coin cell (install '+' side UP).

COMPANION MANUAL CHAPTERS, STANDARDS & KEYSTROKES

BOOK CHAPTERS: SC-32 Reference Manual Ch. 1–14 & Appendices A–E
STANDARDS: Grades 6–12, AP Calculus/Physics, UIL Number Sense & University STEM
WATERLINE SPEC: SC-32 Handheld + Flat-Pack Origami Stand + TPU Sleeve

ALL 3 VERIFIED CALCULATOR LESSON SEQUENCES (SC32):
Ex 1: 3 [↑] [x²] 4 [↑] [x²] [+] [√x] → X=5
Ex 2: 3 [1/x] 4 [1/x] [+] [1/x] → X=1.71428571
Ex 3: 4 [ENTER] 5 [+] 6 [ENTER] 7 [+] [×] → X=117

EQN

BOX ICON KEY: [ENTER], [↑][x²], [1/x] & [r][POLY]
✓ Cross-Referenced to Companion Book & Calculator Engine
Fold back to [01] & keep this reference card inside your box lid!

LOST A PIECE? GRAB SINGLE REPLACEMENTS ANYTIME AT stackcalc.io/parts

✓ KEEP THIS GUIDE IN BOX LID!